



1.INTRODUCTION

The analysis tends to optimize the conditions of redistribution of the learners from the three-secondary schools in Highland City, which followed the closure of a particular school. The redistribution was offered to ensure minimum costs on busing while maintaining both capacity limits and grade-based distributions. Using excel solver, alternatives such as extension of the school capacity using portable classrooms, changing costs, or eliminating short-distance busing, are explored. It then recommends splitting up the residential areas and possibly one more school closure for a more economical allocation. The report describes structured mathematical models that are likely to result in actionable decisions by the council.

2.LITERATURE REVIEW

School district planning seeks to minimize the travel cost of students attending school while adhering to criteria such as the number of students that can be accommodated within the schools, and the respective community sending their children to those schools (Caruso et al., 2012). For instance, development and fairness usually cost travel distance against the community's integrity (Eiselt & Marianov, 2014). Development of modern advanced computation tools like Excel Solver makes it possible to solve very complex problems related to school districts, such as routing buses while assessing environmental conditioning and safety issues (Laporte et al., 2010). When transport savings exceed operational costs, mobile classrooms can implement solutions at relatively low costs for capacity problems (Kim & Lee, 2016). While resource use efficiency is improved by splitting communities, it also complicates the assignment (Ferland & Guenette, 1990). Herein, mathematical programming is utilized to deal with the challenges of the cost, required logistics, and community cohesion taken by Highland city council.

3.METHODOLOGY

3.1 Problem Description

The Highland City Council is working to minimize the cost of busing while still imposing some major constraints. First, students from each residential area must be assigned to one of three secondary schools without leaving any allocation uncovered. Second, the number of students assigned to each school must not exceed its capacity limits. Third, each school must have balanced grade compositions within 30% to 36% of each grade (6th, 7th, and 8th), which satisfies the equality condition across the grades. The assignment should also satisfy the non-negativity condition. These set of constraints make it a costeffective, fair, and feasible solution.

3.2 Mathematical Models

The program has as its intent to minimize the total costs of busing students while satisfying given constraints set on the transportation problem in formulating it as a Linear Programming (LP) problem. The decision variables (x_{ij}) denote the number of students assigned from each area (i) for each school (j). The objective function minimizes total costs, which are defined as the total of busing costs per student (C_{ij}) multiplied by the number of assigned students (x_{ij}). The constraints in the model make sure that all students will be assigned to schools ($N_j = \sum x_{ij} = N$), that school capacities are not exceeded ($N_j = \sum x_{ij} \leq C$) and that the distributions of

grades remain such that they are between 30 and 36 percent of the total populations per school . Each variable is also restricted to being non-negative ($x_{ij} \geq 0$). On these bases, this model will be solved using an operation such as the Simplex LP algorithm which produces optimal student allocation minimizing costs and satisfying all constraints.

3.3 Formulation

The objective is to minimize total busing cost Z , defined as the sum of the products of the number of students assigned and the cost per student for all routes:

$$\text{Minimized } Z = \sum_{i=1}^6 \sum_{j=1}^3 C_{ij} \cdot x_{ij}$$

Linear Constraints:

Constraint 1: Student Allocation: $\sum_{i=1}^6 x_{ij} \leq \text{Total no. of students in area } i$

$$(1) X_{11} + X_{12} + X_{13} = 450$$

$$(2) X_{21} + X_{22} + X_{23} = 600$$

$$(3) X_{31} + X_{32} + X_{33} = 550$$

$$(4) X_{41} + X_{42} + X_{43} = 350$$

$$(5) X_{51} + X_{52} + X_{53} = 500$$

$$(6) X_{61} + X_{62} + X_{63} = 450$$

Constraint 2: School Capacity: $\sum_{i=1}^6 x_{ij} \leq \text{Capacity of School } j, \forall j$

$$(7) X_{11} + X_{21} + X_{31} + X_{41} + X_{51} + X_{61} \leq 900$$

$$(8) X_{12} + X_{22} + X_{32} + X_{42} + X_{52} + X_{62} \leq 1100$$

$$(9) X_{13} + X_{23} + X_{33} + X_{43} + X_{53} + X_{63} \leq 1000$$

Constraint 3: The simplified equations ensure that grade distribution constraints are met by calculating the weighted sum of students for each grade and keeping it within the allowable range, with lower and upper bounds defining permissible deviations.

(10) Equation for Grade 6 of schools:

$$0.30 * \sum_{i=1}^6 x_{ij} \leq 0.32(x_{11} + x_{12} + x_{13}) + 0.37(x_{21} + x_{22} + x_{23}) + 0.30(x_{31} + x_{32} + x_{33}) + 0.28(x_{41} + x_{42} + x_{43}) + 0.39(x_{51} + x_{52} + x_{53}) + 0.34(x_{61} + x_{62} + x_{63}) \leq 0.36 * \sum_{i=1}^6 x_{ij}$$

Lower Bound: $-0.02(x_{11}+x_{12}+x_{13})-0.07(x_{21}+x_{22}+x_{23})+0+0.02(x_{41}+x_{42}+x_{43})-0.09(x_{51}+x_{52}+x_{53})+0.04(x_{61}+x_{62}+x_{63}) \leq 0$

Upper Bound: $-0.04(x_{11}+x_{12}+x_{13})+0.01(x_{21}+x_{22}+x_{23})-0.06(x_{31}+x_{32}+x_{33})-0.08(x_{41}+x_{42}+x_{43})+0.03(x_{51}+x_{52}+x_{53})-0.02(x_{61}+x_{62}+x_{63}) \leq 0$

(11) Equation for Grade 7 of schools:

$$0.3 * \sum_{i=1}^6 x_{ij} \leq 0.38(x_{11}+x_{12}+x_{13}) + 0.28(x_{21}+x_{22}+x_{23}) + 0.32(x_{31}+x_{32}+x_{33}) + 0.40(x_{41}+x_{42}+x_{43}) + 0.34(x_{51}+x_{52}+x_{53}) + 0.28(x_{61}+x_{62}+x_{63}) \leq 0.36 * \sum_{i=1}^6 x_{ij}$$

Lower Bound: $-0.08(x_{11}+x_{12}+x_{13})+0.02(x_{21}+x_{22}+x_{23})-0.02(x_{31}+x_{32}+x_{33})-0.10(x_{41}+x_{42}+x_{43})-0.04(x_{51}+x_{52}+x_{53})+0.02(x_{61}+x_{62}+x_{63}) \leq 0$

Upper Bound: $0.02(x_{11}+x_{12}+x_{13})-0.08(x_{21}+x_{22}+x_{23})-0.04(x_{31}+x_{32}+x_{33})+0.04(x_{41}+x_{42}+x_{43})+0.02(x_{51}+x_{52}+x_{53})-0.08(x_{61}+x_{62}+x_{63}) \leq 0$

(12) Equation for Grade 8 of schools:

$$0.3 * \sum_{i=1}^6 x_{ij} \leq 0.30(x_{11}+x_{12}+x_{13}) + 0.35(x_{21}+x_{22}+x_{23}) + 0.38(x_{31}+x_{32}+x_{33}) + 0.32(x_{41}+x_{42}+x_{43}) + 0.27(x_{51}+x_{52}+x_{53}) + 0.38(x_{61}+x_{62}+x_{63}) \leq 0.36 * \sum_{i=1}^6 x_{ij}$$

Lower Bound: $-0.05(x_{21}+x_{22}+x_{23})-0.08(x_{31}+x_{32}+x_{33})-0.02(x_{41}+x_{42}+x_{43})+0.03(x_{51}+x_{52}+x_{53})+0.08(x_{61}+x_{62}+x_{63}) \leq 0$

Upper Bound: $-0.06(x_{11}+x_{12}+x_{13})-0.01(x_{21}+x_{22}+x_{23})+0.02(x_{31}+x_{32}+x_{33})-0.04(x_{41}+x_{42}+x_{43})-0.09(x_{51}+x_{52}+x_{53})+0.02(x_{61}+x_{62}+x_{63}) \leq 0$

Constraint 4: Non-negativity constraint, $X_{ij} \geq 0$

3.4 Optimization Approach

Data Preparation: Information like busing costs, weights, the absolute number of students in respective areas, schools' total seats, and grade distributions are updated in the Excel sheet format.

Optimum Using Excel Solver: The tool of Excel Solver tries to minimize overall busing costs by solving the objective function with specific constraints. Setting decision variables for an allocation of students to schools. They include school capacity constraints, balance criteria across grades, and nonnegative conditions.

Scenario Analysis: Cancelling Busing for Short-Distance Routes: This exploratory study considers the costs avoided by students who travel less than designated distances, with busing options eliminated.

Portable Classrooms: Are temporary additional classrooms to enlarge a school population made effective in offsetting busing expenses?

Sensitivity Analysis: Changes in the cost of busing will determine how the changed figure will affect the optimized solution, assuring robustness under varying conditions.

Validation and Recommendations: Results are validated for constraint satisfaction like capacity limits and grade distributions.

Optimally interpreted student allocations in terms of areas economically strong and having potential for development, improvement recommendations which are sent to Highland Council, weighing cost savings against the satisfaction of a community while ensuring the solution is effective and fair.

4.OUTPUT AND RESULTS DISCUSSION

(a) Optimal Solution:

In order to find out the optimal solution using the excel solver, we take the coefficient from equation 1 to 15 and enter into constraints 1 to 15. The busing costs from area i to school j are the objective coefficients with all impractical choices are set at £100000 to exclude them from the optimization process.

Cells C3:T3 is the decision variables row.

Total Busing Cost (Z) = £420000

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
2			x11	x12	x13	x21	x22	x23	x31	x32	x33	x41	x42	x43	x51	x52	x53	x61	x62	x63	TOTAL	BUSING COST		
3		Solutio	0	450	0	0	600	0	0	0	550	350	0	0	500	0	0	0	0	450	420000			
4		Obj. Coeffici	300	0	700	1E+06	400	500	600	300	200	200	500	1E+06	0	1E+06	400	500	300	0				
5																					LHS	Sign	RHS	
6		Constra	1	1	1																450	=	450	
7		Constra				1	1	1													600	=	600	
8		Constra							1	1	1										550	=	550	
9		Constra										1	1	1							350	=	350	
10		Constra													1	1	1				500	=	500	
11		Constra																1	1	1	450	=	450	
12		Constra	1			1			1			1			1			1			850	<=	900	
13		Constra		1			1			1			1			1			1		1050	<=	1100	
14		Constra			1			1			1			1			1			1	1000	<=	1000	
15		Grade Distributi																						
16		on 6th Grade Lower Bound	Constra	-0.02	-0.02	-0.02	-0.07	-0.07	-0.07	0	0	0	0.02	0.02	0.02	-0.09	-0.09	-0.09	0.04	0.04	0.04			0
17																					-71	<=		
18		Grade Distributi																						
19		on 6th Grade Upper Bound	Constra	-0.04	-0.04	-0.04	0.01	0.01	0.01	-0.06	-0.06	-0.06	-0.08	-0.08	-0.08	0.03	0.03	0.03	-0.02	-0.02	-0.02			0
20																					-67	<=		
21		Grade Distributi																						
22		on 7th Grade Lower Bound	Constra	-0.08	-0.08	-0.08	0.02	0.02	0.02	-0.02	-0.02	-0.02	-0.1	-0.1	-0.1	-0.04	-0.04	-0.04	0.02	0.02	0.02			0
23																					-81	<=		
24		Grade Distributi																						
25		on 7th Grade Upper Bound	Constra	0.02	0.02	0.02	-0.08	-0.08	-0.08	-0.04	-0.04	-0.04	0.04	0.04	0.04	-0.02	-0.02	-0.02	-0.08	-0.08	-0.08			0
26																					-93	<=		
27		Grade Distributi																						
28		on 8th Grade Lower Bound	Constra	0	0	0	-0.05	-0.05	-0.05	-0.08	-0.08	-0.08	-0.02	-0.02	-0.02	0.03	0.03	0.03	-0.08	-0.08	-0.08			0
29																					-102	<=		
30		Grade Distributi																						
31		on 8th Grade Upper Bound	Constra	-0.06	-0.06	-0.06	-0.01	-0.01	-0.01	0.02	0.02	0.02	-0.04	-0.04	-0.04	-0.09	-0.09	-0.09	0.02	0.02	0.02			0
32																					-72	<=		

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

\$U\$12:\$U\$20 <= \$W\$12:\$W\$20

\$U\$6:\$U\$11 = \$W\$6:\$W\$11

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

(b) The original busing expenditure came to £420,000. The cost for both Option 1 (where students under 1.5 miles do not have to be bussed being £200 per student) and Option 2 (where students under 2 miles are not bused and being £300 or less per student) reduced it to £240,000, giving savings of £180,000 from the original scenario.

The areas of catchment remain unchanged for both options. Students living in areas where the busing cost falls below the thresholds are allocated to other schools instead with an aim at minimizing costs. For example, Area 1 and Area 2 stand for School 2, because that is still the most cost-effective allocation. The same rationale applies for Area 3 and Area 6 to School 3. Area 4 and Area 5 continue to be allocated to School 1, maintaining their optimal assignments unchanged. All schools are within the capacity constraints: School 1 and School 2 have slack for 50 students. School 3 is completely utilized, so there is no remaining capacity.

(c) Evaluating all 3 options:

Scenario	Cost for Busing	Savings v/s Original	Catchment Areas	Constraints Satisfied
Original Scenario	£420,000	£0	Students optimally assigned across schools	Yes
Option 1: <1.5 Miles	£240,000	£180,000	Identical to original	Yes
Option 2: <2 Miles	£240,000	£180,000	Identical to original	Yes

The table represents the three bus rides scenarios effects on costs, savings, and constraints. The Original Scenario represents the baseline, with a total busing cost of £420,000.

Two options besides are Option 1: No busing for students who travel less than 1.5 miles, costing £240,000; and Option 2: No busing for students who travel less than 2 miles outside the catchment area, costing £240,000. Thus, both options save £180,000.

It should also be noted that the catchment is the same for each scenario and for each assignment of students to schools: this means that cost savings achieved in Options 1 and 2 will make no impact on the optimized allocation of students. Also, all scenarios met the required constraints, namely school capacity limit and balanced grade distribution making them feasible. It shows that Option 1 or Option 2 can be exercised without much doubt as both achieve great savings at the same time without jeopardizing the optimized student distribution, violating constraints.

(d)

Factors	Busing Cost and Allocation
Current Route Costs	No learners assigned to x61; no impact on total cost or allocation.
Cost of Route x61 Falls Below £500	Route x61 becomes cost-effective; allocation from Area 6 may change.
10% Increase in Costs to Area 6	Likely no change if current routes remain cheapest.
Significant Cost Increase to Area 6	May require reassignments to minimize costs while meeting constraints.

In the current optimum solution, there are not yet any learners assigned to the route between Area 6 and Secondary School 1 ($x_{61}=0$). This means that an increase in the cost of this route would not affect the total busing cost or student allocation. However, should the costs of the route fall below £500, it may become a cost-effective option, possibly modifying the allocation of students from Area 6. A 10% increase in the busing costs for all routes to Area 6 might not change the existing allocation if the existing routes remain the cheapest; however, much higher cost increases may compel reassignments to minimize cost while satisfying all the

other very important constraints-both school capacities and balanced grade distributions. Thus, the present solution demonstrates flexibility and robustness in adjusting to cost changes while still reaching the best possible efficiency and meeting all requirements.

(e) Additional portable classrooms, costing an annual amount of £2,500 with a maximum capacity of 20 students costs would only work if it could save more from busing than the costs incurred for the classrooms being a viable solution to cost-effectively increasing school capacity and thereby reducing busing cost. Students from Area 6 are assigned to School 3 with no busing cost associated with this route; therefore, these portable classrooms are not needed under present conditions. However, should the busing costs for routes to School 1 or to other schools significantly reduced, then perhaps it may now be cost-effective to reallocate students and install portable classrooms at School 1 or School 2. Further, increasing capacity at School 3 may optimize allocations from higher cost areas like Areas 5 and 6. The council should also closely monitor changes in busing costs and capacity demands so that they may reassess the usefulness of these portable classrooms should there be opportunities for reallocation warranting the investment.

Scenario	Cost of Portable Classroom	Capacity Added	Impact on Solution	Recommendations
Current Solution	£2,500 / classroom	20	No impact as students from Area 6 are assigned to School 3 with no busing cost	No portable classrooms needed
Cost Decrease for Area 6 to School 1	£2,500 / classroom	20	Could reallocate to School 1 if busing costs decrease significantly	Monitor costs; recheck if reallocation becomes cost-effective
Capacity Increase at School 2	£2,500 / classroom	20	Could absorb students from higher cost routes to School 1 or School 3	Explore if busing cost reductions justify the investment
Capacity Increase at School 3	£2,500 / classroom	20	Could optimize allocations from <u>high cost</u> areas like Areas 5 and 6	Consider if cost savings exceeds the investment

(f)The system currently best for economically distributing and maintaining all residential areas within one catchment zone is placing every student from every residential area in one school. This prevents the logistical and disturbance problems that may arise from putting children in different schools by minimizing the total cost of busing beyond GBP 500,000.

(g) In an event where any one of the three secondary schools is closed, the capacity of the remaining two schools will increase by 500 each leading to a student reassignment according to grade distribution capacity requirements plus reduction in busing expenses.

Closing School 1 is the cheapest option. Being more proximate to the impacted areas, Schools 2 and 3 can manage more efficiently the relocated pupils and therefore cause the lowest increase in busing costs.

This considered, closing School 2 or School 3 would incur significant cost increases since the students attending these schools would travel farther distances to reach the other schools. Thus, closing School 1 is the most logical suggestion for the Highland City Council as it brings in cutting costs and logistical problems. Further confirmation with revised cost data is recommended to finalize this decision.

School to be closed	Reallocation Impact	Expected Busing Costs	Feasibility	Recommendation
School 1	Students reallocated to Schools 2 and 3	Lowest additional cost	Efficient reallocation due to proximity	Recommended to close
School 2	Students reallocated to Schools 1 and 3	Higher costs due to longer distances	Challenging due to higher student capacity	Not recommended
School 3	Students reallocated to Schools 1 and 2	Highest additional cost	Significant logistical challenges	Not recommended

- If school 1 is closed, total busing cost is GBP 725,000.
- If school 2 is closed, total busing cost is GBP 655,000.
- If school 3 is closed, total busing cost is GBP 700,000.

School 1 is the most cost-effective closure option due to efficient reallocation and minimal increase in busing costs.

Closing **School 2** or **School 3** leads to higher costs and greater logistical challenges making them less feasible.

5.RECOMMENDATION

Minimize Total Busing Costs: Keeping an eye on busing expenses, consider portable classrooms only where the savings achieved through reduced transportation cost greatly exceed the cost of classroom leasing.

Evaluate Short-Distance Busing Options: With respect to short-distance busing, it should either be Option 1 or 2, and go for Option 1 to save students inconvenienced and money spent.

Future Planning: Reduce the need for buses while achieving sustainability through land use guidelines and investment in biking and pedestrian infrastructure.

Enhance Efficiency and Monitor Trends: Regularly review and optimize bus routes for cost savings with travel time reduction. Maintain an eye on enrolment and changing demographics to reposition schools and their resources proactively for long-term effectiveness.

6.Conclusion

The analysis suggests that busing costs should be cross-optimized to £420,000 by either continuing current allocations or adopting Option 1 (busing will be eliminated for routes shorter than 1.5 miles) and saving £180,000. If busing costs escalate exorbitantly, there can be an argument for portable classrooms, but splitting residential areas is kept out from restrictions so that costs will not rise above £500,000. For school closure, School 1 is the preferred choice because it would be the least expensive. These are strategies for balancing cost savings along with logistical viability against community needs.